

Education

New York University, Courant Institute
Ph.D. in Computer Science
Advisors: Gautam Kamath and Anupam Gupta

New York, NY
2026 – Present

Columbia University
Bachelor of Science, Applied Mathematics, Minor in Computer Science
Magna cum laude, GPA: 4.02

New York, NY
2022 – 2026

Publications

Szymon Snoeck, Noah Bergam, and Nakul Verma. Compressibility Barriers to Neighborhood-Preserving Data Visualizations. International Conference on Algorithmic Learning Theory (ALT). 2026

Noah Bergam, **Szymon Snoeck**, and Nakul Verma. t-SNE Exaggerates Clusters, Provably. International Conference on Learning Representations (ICLR). 2026

Manuscripts

Szymon Snoeck. A Uniform Convergence Result for Learning Text Data. 2025

Szymon Snoeck. The Negative Inter-Dependencies of the Multivariate Hypergeometric Distribution. 2025

Szymon Snoeck, Christopher En, Yuri Faenza. The Difficulty of Approximating Nash Social Welfare in Online Matching. 2024

Szymon Snoeck, Sam Wang. Deterministic Approximate Counting $\mathbb{F}_2$ Polynomials Via Correlation-based Fourier Bounds. 2024

Talks

Compressibility Barriers to Neighborhood-Preserving Data Visualizations | ALT

Spring 2026

Research Experience

Research Project with Prof. Alexandr Andoni
Fine-Grained Complexity and Algorithms

Columbia University, New York, NY
Summer 2025 – Present

- Studied data-dependent lower bounds for nearest neighbor search and worst-case to average-case tradeoffs
- Currently working towards a publication on approximate online matrix-vector multiplication

Research Project with Prof. Nakul Verma
Dimension Reduction Research

Columbia University, New York, NY
Spring 2023 – Summer 2026

- Explored the shortcomings of data visualization methods, including t-SNE and UMAP, from a theory perspective

Research Project with Prof. Yuri Faenza
Online Matching Theory Research

Columbia University, New York, NY
Summer 2024

- Studied the limits of online bipartite matching under various notions of welfare

Teaching Experience

Computer Science Department

Columbia University, New York, NY

Teaching Assistant, COMS 4771: Machine Learning Spring 2024, Fall 2024, Spring 2025, Summer 2026 (4 terms)
Teaching Assistant, COMS 4774: Unsupervised Learning Fall 2025
Teaching Assistant, COMS 4232: Advanced Algorithms Spring 2026

Service

Referee, Nature Communications Spring 2026
Volunteer, ICLR Spring 2026

Awards

NSF Graduate Research Fellowship | National Science Foundation Fall 2026 – Spring 2029
Undergraduate Applied Mathematics Faculty Award | Columbia University Spring 2026
Awarded to one graduating applied mathematics student each year for their outstanding academic achievements
CRA Outstanding Undergraduate Researcher Award Finalist | Computing Research Association Fall 2025
Dean’s List | Columbia University Fall 2022 – Spring 2026

Grants and Funding

SUMMER@SEAS Research Grant, Columbia SEAS. *Is Data Visualization Trustworthy?* \$6,000 Summer 2025
SUMMER@SEAS Research Grant, Columbia SEAS. *Fairer Online Matching.* \$6,000 Summer 2024

Skills

Technical: Proficiency with $\text{\LaTeX}$, Java, Python, C, scikit-learn, NumPy, TensorFlow, and Pandas
Language: Dutch (Fluent) and French (Elementary Proficiency)
Interests: Photography (work exhibited at the Metropolitan Museum of Art), inking, competitive debate